

3(a)	<i>Mycobacterium tuberculosis</i> / <i>Mycobacterium bovis</i> ; correct spelling	1
3(b)	1 deaths decrease to zero / new cases fall to 4 per 100 000, (in 2010) ; A other values that confirm decrease <i>(after introduction of antibiotics)</i> 2 deaths relatively constant / approx. 52 per 100 000, until 1943–6 (<i>accept any in range</i>) ; A for a few years A idea that rate of decrease not improved 3 number of new cases initially stops increasing / plateaus ; 4 (overall) number of new cases increases before decreases ; 5 data to support mp 3 or 4 ; e.g. 88–90 per 100 000 (1940–44) rises to / drops from, 104 per 100 000 in 1946 ; <i>(after introduction of vaccination)</i> 6 no change in trend of decrease in deaths / deaths continue to decrease ; 7 decrease from 36–40 per 100 000 (in 1948) ; 8 short plateau / 100 per 100 000 (until 1950), in new cases, then decreases ; 9 antibiotics may not have been in widespread use at first ; ora 10 vaccine, reduces spread / gives (herd) immunity ; 11 <i>ref. to</i> decrease less steep initially as time needed to build up herd immunity ; 12 AVP ; e.g. suggestion why increase in new cases after introduction of antibiotics suggestion why cases have not decreased to zero not possible to tell relative effect of vaccine v antibiotic	max 5

Question	Answer	Marks
3(c)(i)	some people who are HIV+ will have developed HIV / AIDS ; <u>HIV / AIDS</u> weakens <u>immune</u>, system / response ; ora <i>allow ecf for HIV+ people</i> detail of why immune response weakened ; people with HIV / AIDS, prone to opportunistic diseases ; A TB is an opportunistic disease A more susceptible to, diseases / TB (Table 3.1 shows that) a greater proportion / AW, people die of TB if they are HIV+ ; AW e.g. greater chance of dying from TB if HIV+ dormant TB more likely to become the active form in, HIV+ people / people with HIV / AIDS ; AW AVP ; a high proportion of deaths from HIV / AIDs is due to TB mortality HIV+ people do not respond well to treatment for life-threatening diseases inability to pay for treatment for all conditions for HIV+ people	max 3
	Total:	9